

Bug Report Classification by Selecting Relevant Features Using Chi Square, Information Gain and Latent Semantic Analysis

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Abstract— In machine learning, feature selection is a very important step to reduce the dimensionality of data by removing irrelevant features, redundant data to improve the learning accuracy. As the dimensionality of data has increased, feature selection has become a challenging task. Various approaches have been proposed for feature selection. In this study, we have analyzed the effectiveness of three widely used feature selection methods namely Chi square; information gain and latent semantic analysis (LSA) to classify the software bugs. The performance of four classifiers K nearest neighbor, Random Forest, naïve bayes and support vector machine are evaluated for the above feature selection methods in terms of accuracy, precision and recall.

Keywords— Feature selection, Bug Report Classification, Chi square, Information gain, Latent Semantic Analysis

I. INTRODUCTION

Feature selection is the process of selecting a minimum subset of features from the original feature set, which contribute most to a classification task. It is used to reduce the dimensionality of the feature set by removing irrelevant, noisy and redundant data [1-4]. Feature selection allows classifiers to train effectively with relevant feature set. In general, features can be categorized into irrelevant, relevant and redundant. The aim of the feature selection methods is to select the relevant features for solving a particular problem to increase the performance of a classification model. Feature selection methods are used in various applications such as data mining, signal processing, and image processing. Feature selection methods can be developed according to different statistical techniques [5, 6]. Three methods namely; filter [7] embedded model and wrapper exist in the literature. Filter methods are based on the relationship between the features and the category label. Chi square and information gain are the examples of filter method. The embedded model methods select the features at the time of training. The examples of embedded methods are ID3 and C4.5 which are based on decision trees. Wrapper methods are based on search strategy and evaluate each subset for selecting the relevant features. This process is repeated until relevant subset of features is selected. These methods are computationally expensive than filter methods [8]. The effectiveness of these feature selection algorithms is evaluated by analyzing the performance of classifiers. In this paper, the performance of four classifiers K nearest neighbor, Random forest, Naïve bayes and Support vector machine are evaluated for chi square, information gain and latent semantic analysis feature selection methods in terms of accuracy, precision and recall.

The rest of the paper is organized as follows: section II discusses the related work based on the feature selection techniques. Section III presents the proposed framework. Section IV explains the feature selection methods used in this study. Classifier and evaluation metric are discussed in section V. The dataset and results are also discussed in this section. The section VI presents all the threats that can affect the performance of classifiers. Section VII concludes the study with future work.

II. RELATED WORK

There are many methods exist in literature to select the relevant features. These methods can be categorized into two types: supervised and semi supervised feature selection methods. Supervised feature selection methods use for labelled data such as fisher score [9] which rank features according to fisher criterion. It selects features independently based on their score and results in an optimal subset of features. It does not consider the correlation between the features. Linear discriminant analysis (LDA) [10] is a generalization of Fisher's linear discriminant and was proposed to find out the combination of features that which best explain the data by maximizing the ratio between and within the classes. However, LDA has its own limitation as it suffers from a small sample size i.e. the number of training samples is less than the dimension of the data. To overcome the problem of LDA, Maximum margin criterion [11] based algorithms was proposed which uses a linear combination of features between and within the classes. All the supervised feature selection methods suffer from the unavailability of sufficient labelled data and performance of these methods also affected.

Semi-supervised feature selection methods use labeled and unlabeled data. It has the advantage over supervised methods as these methods are able to select features form unlabeled data. In the literature semi-supervised Discriminant Analysis (SDA) [12] was proposed which can use the limited number of labelled data and large number of unlabeled data for training. It calculates the class indicator matrix of the training data and assumes that most data lies in low dimension. However, it is usually less efficient on handling large-scale data. It considers a very hard problem due to the absence of labelled data and needs to study further.

III. PROPOSED FRAMEWORK

The overall framework of our approach is depicted in fig 1. The framework consists of six steps: Bug reports collection, Text pre-processing, Tokenization TF-IDF,

Feature Selection and Classification. Each step is described in detail.

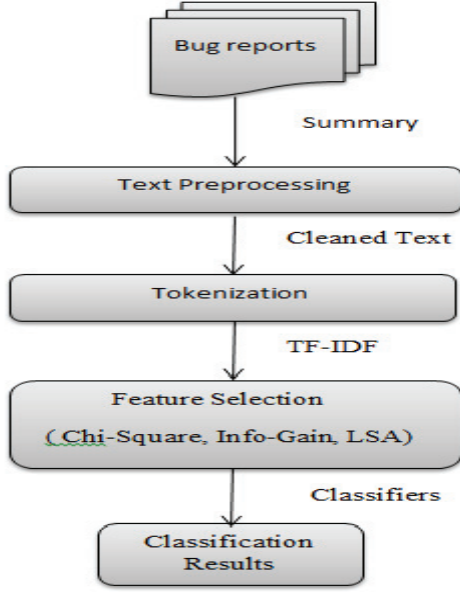


Fig. 1. Proposed Framework

A. Bug report classification

We have collected bug reports from a bug tracking system called MantisBT. Fig 2 shows the snapshot of bug reports from MantisBT. It contains various fields ID, number of comments, attachment, category, severity, status, updated and summary. For this study we have used only

Category	Severity	Status	Updated	Summary
installation	text	resolved (dregad)	2021-08-27	Issue on Ms Edge/IE11
other	minor	resolved (community)	2021-08-25	function gpc_set_cookie() ignores \$p_httponly argument
administration	minor	resolved (dregad)	2021-08-24	Do not buffer output for CLI scripts
upgrade	major	resolved (dregad)	2021-08-24	Improve handling of unserialize->json conversion during upgrade
tools	feature	resolved (dregad)	2021-08-24	New build script to download updated font files

Fig. 2. snap shot of Mantis bt

two fields summary and category. The bug reports are categorized into 49 categories but only six categories namely feature, bug tracker, administration, email, customization and lters are considered. Rest of the categories does not have the sufficient bug reports.

B. Bug report classification

A bug report contains various fields like bug ID, Severity, Priority, Summary, OS, keywords, number of comments, status etc. The summary is short description about a bug. The text in the summary field is of unstructured form which needs to be preprocessed and cleaned. Standard text preprocessing techniques are applied to clean the text in the summary field. The text is preprocessed and cleaned by converting into lowercase, removing punctuation, special symbols and stop words. The stop words are provided in the NLTK library.

C. Tokenization

Tokenization is the process of converting the text into tokens. After the text is pre-processed and cleaned, each word from the summary is separated.

The TF-IDF helps us to understand the importance of a term in a given document.

D. TF-IDF

The TF (Term Frequency) calculates the occurrence of given word in the entire document and IDF (Inverse document frequency) calculates the ratio between the total document and the number of documents containing the given word [13]. The whole idea is to assign higher weights to the infrequent word in the corpus.

E. Feature Selection

Three well known feature selection methods namely feature selection methods Chi square; information gain and latent semantic analysis are used to select the relevant features.

F. Classification

In this study, bug reports are classified into six categories using K nearest neighbor(kNN), Random Forest(RF), naïve bayes (NB) and support vector machine (SVM) algorithms . Their performance is measured in terms of accuracy, precision, recall and f1 score.

IV. FEATURE SELECTION METHODS

The Chi-squared (χ^2) feature selection is measured to test the independence between feature and target. It simply evaluates the significance of each of the features. Chi-square score is calculated using equation (1) between the feature and target variable

$$\chi^2 = \frac{\sum(O_f - E_f)^2}{E_f} \quad (1)$$

where O_f and E_f refers to the actual and expected number of class observations respectively. The expected number of class observations E_f would be expected if there were no relationship between the feature and the target variable. The chi squared requires numeric features to be discretized before calculating so the sum is over each value of the feature is calculated. Most relevant features are selected with high chi square score.

Information gain (IG) is a feature selection method used in decision tress by calculating the gain of each variable in context of the target variable. It calculates the mutual information between the variables. It provides a way to use entropy to know how a change in dataset affects the purity i.e. class distribution. The smaller value of entropy suggests more purity in the dataset. The limitation of IG is the biasness towards the features having more values.

Latent semantic analysis (LSA) is another feature selection method for computer modeling and simulation of the meaning of words by analyzing the text corpora. It closely approximates many aspects of natural language by inferring the semantic relation of words in a text document. It utilizes singular value decomposition by keeping k singular values and hence reduces the dimensionality of the data.

V. CLASSIFIERS

The k nearest neighbors is a non-parametric classification method. It is used in various fields such as information retrieval, data mining and machine

learning[14]. It is also known as lazy learner [15] due to its training nature as, it is just the storage of labeled data. Given a document 'd' with unknown class, a parameter k and a set of documents D, it computes the nearest document for 'd' based on a distance function such as the Euclidean distance function. To find an optimal value for parameter k is challenging task but by experimenting and using various heuristic methods, the best 'k' can be selected.

Naïve Bayes is widely used in text classification problem [16]. Naïve bayes assumes the values of the features are independent to each other. For text classification problem, it assumes that the probability of a word in document is independent of the other words belong to the same document. There are two types of text classifier based on naïve bayes; multivariate Bernoulli NB and Multinomial NB. Multivariate Bernoulli NB represents document as a binary vector based on the presence and absence of a word in the document whereas Multinomial NB considers the word frequency in the document [17]. Random Forest (RF) classifier contains a set of random trees [18]. The term random refers to the selecting any feature to build the individual trees. Bagging method [19] is used during training the classifier, in which individual trees are trained in parallel based on the available features. Bootstrapping aggregation ensures that the every tree in the random forest is unique and hence reduces the variance of the classifier. It is also able to classify large data with high accuracy. Due to its random nature, it is less prone to over fitting. Random forest produces more stable results as compare to other machine learning algorithms.

Support Vector machine (SVM) is most widely used supervised classification algorithm which was introduced by [20]. It is used in various application such object recognition, image and text classification. It is a binary classifier that separates two classes by finding the optimal hyper plane by the maximum margin between the data points. SVM can be extended to a multiclass classification problem by combining many binary SVM classifiers. The strengths of SVM are its effectiveness on high dimensional data, less memory consumption, and large datasets due to variety of kernel functions used.

A. Datasets

Two datasets namely Cassandra and MantisBT are used to implement and analyze the effectiveness of feature selection methods. We have taken the same datasets as [21]. Table I shows the details of the datasets. A Bug tracking system called mantis built on PHP was used to collect the bugs from <https://www.mantisbt.org/bugs> containing bugs from 2005 to June 2021. The dataset consists of 1310 bug reports classified into 49 categories.

TABLE I
DETAILS OF DATASETS

Project	Size	Category Label	Source
Cassandra	1618	10	http://odc.dei.uc.pt
MantisBT	1310	49	https://www.mantisbt.org/bugs

The category label represents the number of categories assigned to label the defect reports. We consider only 6 categories to classify the bugs. The distributions of categories are shown in Table II.

TABLE III
BUG DISTRIBUTION

Cassandra		MantisBT	
Capability	557	feature	198
Reliability	309	bugtracker	173
Requirements	70	administration	104
Integrity/Security	43	email	83
Performance	36	customization	81
Serviceability	32	lters	73

B. Evaluation Metric

In this study, we used accuracy, precision and recall to evaluate the performance of the kNN, NB, RF and SVM classifiers. The metrics accuracy, precision and recall are calculated based on the confusion matrix. It is a tabular presentation of the performance of the model prediction. Confusion matrix compares the values of the predictive class to the actual class values. Consider a given class the following measurements can be taken:

True Positive (TP): refers to the number of instances correctly classified as class C.

False positives (FP): refers to the number of instances classified incorrectly as to another class.

True negatives (TN): refers to the number of instances not classified as C.

False negatives (FN): refers to the number of instances assigned to other classes but they actually belong to class C.

The evaluation metrics accuracy (Acc), Precision (Pre)and Recall (Rec) now can be defined as :

$$acc = \frac{TP+TN}{TP+FP+FN+TN} \quad (2)$$

$$pre = \frac{TP}{TP+FP} \quad (3)$$

$$rec = \frac{TP}{TP+FN} \quad (4)$$

C. Results and discussion

Effectiveness of feature selection methods namely Chi square; information gain and latent semantic analysis is evaluated using two datasets: Cassandra and MantisBT. The performance of classifiers kNN, NB,RF and SVM on

TABLE III.
PERFORMANCE OF CLASSIFIERS ON CASSANDRA DATASET

Feature selection Method	Classifiers	acc	pre	rec
Chi-Square	kNN	.5047	.2849	.1651
	NB	.5000	.1178	.1645
	RF	.4857	.1250	.1620
	SVM	.5142	.2519	.1692
Info-gain	kNN	.5190	.2524	.1729
	NB	.5142	.2519	.1692
	RF	.4714	.2108	.1971
	SVM	.5095	.3863	.1666
LSA	kNN	.5047	.1274	.1672
	NB	.4932	.2410	.2063
	RF	.5095	.1834	.1721
	SVM	.4547	.1250	.1125

Cassandra dataset are shown in table III. For chi-square, info gain and LSA, the best accuracy value are represented in **bold**.

The accuracy obtained with reducing the feature set by info gain is 51% for kNN classifier which is the highest value as

TABLE IV.
PERFORMANCE OF CLASSIFIERS ON MANTIS BT DATASET

Feature selection Method	Classifiers	acc	pre	rec
Chi-Square	kNN	.5657	.4428	.1666
	NB	.5335	.4512	.4427
	RF	.4986	.4465	.3692
	SVM	.5195	.4814	.3572
Info-gain	kNN	.4936	.4393	.2962
	NB	.4405	.4506	.4336
	RF	.4055	.4407	.3920
	SVM	.5356	.5104	.2780
LSA	kNN	.3566	.5248	.3451
	NB	.5055	.4978	.3246
	RF	.4021	.3462	.3019
	SVM	.3496	.3958	.2874

compare to other feature selection methods. The results can be improved by collecting a large quantity of data to classify bug reports in the 6 categories. Certainly, we have a scope for further improvement. Table IV displays the performance of the classifiers in terms of accuracy, precision and recall for mentis dataset. Regarding the mentis bug reports, kNN, SVM and NB were able to classify with accuracy of 56%, 53% and 50% when selecting the relevant features by chi-square, Info gain and latent semantic analysis respectively. Still it is important to mention that we are using the six categories as we do not have sufficient data for rest of the categories. Chi-square feature selection technique extracts the more relevant features from the studied datasets as compared to other two used in the study.

VI. THREATS TO VALIDITY

In this section, we identify the external, construct and internal validity threats that might appear in our study.

Concerning the external threats, an obvious threat is imbalance distribution of the classes in the dataset. As a result, there is high chance of biased learning. We have tried to remove the biasness by taking random samples of varied distribution. The size of dataset and summary of each bug is also a concern to the external threat. In the future, this threat can be mitigated by taking a larger dataset to make the results more accurate.

Concerning construct validity, only the summary attribute is considered for all the bugs in the dataset. Apart from summary, more attributes about the bug could be extracted. The dataset is converted into 6 class problem because of the insufficient bug reports in the remaining classes. Further, the standard natural language processing techniques are applied to preprocess the text in the summary field to improve the classification.

The internal validity is concerned with representation of the studied systems and performance of the classifiers on

dataset. However, the text provided by users while reporting the bug is not in a structured form.

VII. CONCLUSION AND FUTURE WORK

This study evaluated the impact of feature selection methods Chi-square, information gain and LSA to classify the bug reports from two project; Cassandra and MentisBT. Four widely used machine learning algorithms: k-Nearest Neighbors, Naïve Bayes, Random Forest and Support Vector machine to compare the effectiveness of feature selection algorithms in terms of accuracy, precision and recall. The experimental results show that these feature selection methods affects the classification performance of the classifiers. It has been analyzed that there are number of factors which can affect the performance such as size, diversity in projects, and textual description of the defect reports. For the future work, these factors must be considered to improve the performance.

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